Nisha Nisha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.57
Excellent Pre University College Sunnari	Class XII (Karnataka School Examination and Assessment Board)	2024	98.83%
Government High School Kalavara	Class X (Karnataka Secondary Education Examination Board)	2022	100.00%

PROJECTS

Bank Account Transaction History System

- Developed a high performance console based financial ledger application in C++ designed to process continuous, real time deposit and withdrawal entries while dynamically updating account balances.
- Implemented memory efficient dynamic array vectors to maintain a persistent, sequential transaction log designed for high throughput entry tracking and ledger history auditing.
- Designed low level algorithmic utilities completely from scratch, including an O(n) Linear Search algorithm for targeted ID/date transaction lookup and a customized two pass Bubble Sort routine for multi criteria chronological and value based financial data sorting.

ESSENTIAL COMMODITY PRICE SPIKE & EARLY-WARNING MODEL:

- Built a predictive machine learning pipeline using Python (Pandas, NumPy, Scikit-learn) on WFP food price data (more than 10,000 historical records) to monitor retail and wholesale commodity volatility across Indian markets.
- Engineered a feature-preprocessing framework encoding spatial (State, District, Mandi, Lat/Lon) and temporal (Year, Month) metrics to train a Gradient Boosting Regressor, establishing baseline evaluation metrics (R^2, MAE, RMSE).
- Extracted model feature importances (feature_importances_) to isolate primary macro price drivers, identifying commodity categorization, market location parameters, and seasonality as key factors in price variance.
- Developed an automated diagnostic reporting engine that maps model predictions back to regional administrative zones, identifying and ranking the top 10 highest commodity price surge risks.
- Translated algorithmic price forecasts into actionable policy insights for governance consultants, enabling administrative leadership to prioritize targeted buffer-stock releases and open-market supply interventions.

Electronic Nose: Gas Sensor Data Analysis

- Engineered a robust end to end Python data preprocessing pipeline utilizing Pandas and Scikit-learn to systematically ingest, clean, and align complex sequential data streams from the UCI Gas Sensor Drift Dataset.
- Mitigated extreme multi scale sensor volatility across temporal batches by executing standard feature normalization and evaluating non linear feature redundancy via detailed correlation matrix mappings to diagnose heavy multicollinearity.
- Conducted dimensionality reduction using Principal Component Analysis to compress 120+ high dimensional sensor features into an interpretable 2D visual space, successfully mapping long term temporal drift trajectories and tracking array degradation patterns across 10 chronological batches.

SKILLS

Languages : *Python* | *C* | *C++* | *SQL*

Libraries and Tools : *Solid Edge* | *NumPy* | *Pandas* | *Scikit-learn* | *Matplotlib* | *PyTorch* | *Git* | *Jupyter Notebook* | *VS Code*

RELEVANT COURSES

Mathematics courses : Calculus-1, Calculus-2, Differential Equations, Elementary Linear Algebra, Applied Mathematics in Chemical Engineering, Numerical Methods.

Programming and AI related courses : Introduction to Programming, Artificial Intelligence.

POSITIONS OF RESPONSIBILITY

Operations Coordinator, Finance & Consulting Club:

- **Workshop Operations & Execution:** End to end management of 3 flagship workshops centered on finance and management consulting, driving participant engagement and operational logistics for more than 500 attendees each.
- **Corporate Liaison:** Served as the primary Point of Contact for senior representative from **UTI Mutual Fund**; facilitated session planning, speaker accommodations, and content delivery for a high-impact financial literacy workshop.
- **Hackathon Coordination:** Spearheaded core logistics and event management for a consulting hackathon, overseeing schedule management, judge coordination, and live troubleshooting for 50+ competing teams.

SENIOR UNDER OFFICER (SUO) , NCC:

- **Leadership & Promotion:** Promoted to Senior Under Officer (SUO), the senior-most NCC cadet rank, after three years of progressive leadership roles, culminating in currently leading and mentoring a unit of **50+ cadets.**
- **Financial Administration:** Managed financial record-keeping and prepared balance sheets for unit expenditures and funds, liaising directly with the Director to resolve budgeting and allocation issues.
- **Training & Mentorship:** Delivered structured training sessions for junior cadets, helping them to understand the purpose of NCC building hands-on instructional design and mentorship experience across multiple cohorts.
- **Community & Social Service:** Planned and executed large-scale social-service campaigns including plantation drives, Clean India drives, and blood donation camps, driving cadet and IITH community participation in eco-friendly and civic initiatives.

EXTRACURRICULAR

Have earned Black Belt (1st Dan) in Karate after 10+ years of rigorous training since childhood in 2021.

Led the IIT Hyderabad Women's Athletics Team as the captain, representing the institute at the Inter IIT Sports Meet.

Practiced Yakshagana, a traditional Karnataka theatre art form for over 8 years and performed about 14 stage roles.